

Pavan Kumar Etta

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SKILLS

Languages: Python, Go, Java, JavaScript, TypeScript, C, C++, SQL, C#, Scala, Kotlin, R, .NET, Rust

Backend & Testing: Django, Spring Boot, Node.js, FastAPI, Express.js, gRPC, Test-Driven Development (TDD), Selenium, JUnit

Databases & Tools: AWS, GCP, Azure, Redis, Prometheus, PostgreSQL, MongoDB, Elasticsearch, Splunk, Datadog, Grafana, Docker

Web Technologies: HTML, CSS, React.js, React Native, Angular.js, Next.js, Vue.js, GraphQL, Flask, Redux, REST API, CI/CD, Git

Frameworks: NumPy, Pandas, scikit-learn, PySpark, Hadoop, NLTK, Swift, OpenCV, PyTorch, TensorFlow, Agile, Microservices

EXPERIENCE

University Student Researcher

Nov 2025 – Present

University of Massachusetts — Hadoop, BERT, PySpark, AWS, Snowflake, Airflow, PyTorch

North Dartmouth, MA

- Architected a **Next.js**, **Flask**, and **MongoDB** application for uploading financial documents and generating detailed NLP-powered reports, using **LangChain** to coordinate document analysis, information extraction.
- Designed and built a production document-routing service using **Python**, **BERT**, and **PyTorch**, achieving **93%** accuracy through input validation, error handling, automated classification, and routing documents to specialized report-generation pipelines.
- Built a **TypeScript** framework for pipeline configuration and monitoring, supporting an event-driven **AWS Glue**, Step Functions, **S3**, and Athena architecture that replaced **20 GB Hadoop** batches and reduced latency by **50%**.
- Managed deployments on **Linux-based** servers using **Docker** and facilitated public accessibility by hosting the services.
- Reduced financial-report query time by **60%** by optimizing Snowflake models and automating reliable end-to-end dbt-Airflow data.

Software Engineer, Backend Engineer

Sep 2022 – Jul 2024

Rhenix Lifesciences LLP — Java, Spring Boot, Go, Kafka, Kubernetes, PostgreSQL

Telangana, India

- Engineered **Java** and **Spring Boot** services on Linux for five wearable providers, using OAuth 2.0, **TCP/IP-based APIs**, **Kafka**, **PostgreSQL**, and **EC2** to deliver reliable device-to-cloud health data.
- Reduced data-transformation code by **60%** by building a **C#** and **.NET** backed by **Azure Bus Service** that standardized activity, heart rate, sleep, and SpO2 data from multiple wearable providers enabling consistent monitoring for diabetic patients.
- Led correlation of wearable and **EMR** data to identify heart-rate and sleep patterns across **5+** clinical use cases, doubling efficiency through **Go services**, reliable **gRPC** communication, **Kubernetes** scaling, and **CockroachDB** data management.
- Built a production **RAG** system ingesting **Slack** and **Zoom** conversations, indexing content in a **S3** vector knowledge base, enabling embedding-based semantic retrieval, and generating **Amazon Bedrock** responses with **sub-200 ms** ingestion latency.
- Built a wearable-data analytics platform that processed **10,000s of events per day** from **10+** microservices using **Apache Spark**, **Parquet**, batch and streaming pipelines, and **Elasticsearch** for real-time analytics.
- Conducted performance testing of APIs using **Postman** and **JMeter**. Performed service analysis with Google Analytics and utilized **Grafana** and **Prometheus** for metric analysis to optimize system efficiency and enhance backend services.
- Prevented data loss by extending backup and recovery support for offline wearables, **SAP HANA**, and **Oracle RAC**, enabling record replay after reconnection, eliminating **100+** sync failures, and improving delivery success by **30%**.
- Spearheaded **Testim** and **Selenium** regression automation, increasing testing speed **80%**, while deploying Dockerized backend services through GitHub Actions **CI/CD** pipelines on **Google Kubernetes Engine**.

Software Engineer Intern

Dec 2021 – Jul 2022

CureScience Institute — Android, BLE, REST APIs, MQTT, Lambda, EventBridge

San Diego, CA

- Enabled **24/7** asset tracking and reduced battery consumption **85%** by optimizing **Android BLE** scanning, routing updates through REST APIs, processing them with Lambda and **EventBridge**, and powering Django analytics from DynamoDB.
- Reduced unit-entry errors by **30%** by building **Flutter** mobile applications and **Angular** and **JavaScript** web components, validating **XML** and **HTTP** data exchanges to automatically adjust measurements based on regional settings.
- Minimized gateway-failure detection from **15 minutes to under 1 minute** by instrumenting **MQTT** and **Kinesis** data flows, monitoring API CRUD errors and message-processing latency with CloudWatch, and triggering **SNS** alerts for anomalies.

PROJECTS

Fault-Tolerant Sharded Key-Value Store — C++, Raft, SQLite

- Engineered a fault-tolerant sharded key-value store in **C++** using **Raft** consensus, **SQLite**, leader election, log replication, and replicated state machines to maintain consistency and support recovery from server failures.

Real-Time Mentor Matching Platform — Node.js, Next.js, Socket.IO, Docker, PostgreSQL, AWS EC2, RDS, S3

- Enabled mentors and mentees to connect instantly through a **Node.js** and **Next.js** platform, scaling **Socket.IO** messaging on **EC2** and **PostgreSQL RDS** to 300 concurrent users, 10,000+ messages, 99.8% delivery, and sub-200 ms p95 latency.

EDUCATION

University of Massachusetts, Dartmouth

North Dartmouth, USA

M.S. in Computer Information Science, Data Science, GPA : 4.0

Aug 2024 – May 2026